Aaron Broukhim

Email: aabroukh@ucsd.edu **Mobile:** (310) 867-3564
Website: https://aabroukh.github.io **LinkedIn:** linkedin.com/in/aabroukh

Experience

June 2026 to Sept 2026

Adobe - *Machine Learning Engineering Intern (Incoming)*
• Joining the Digital Experience AI & Product Engineering team to build ML/AI systems for marketing automation, including AI agents and reinforcement learning applications.

Sep 2022 to June 2027

University of California San Diego - *Graduate Researcher, PhD in CSE (Advisor: Nadir Weibel)*
Thesis: preference-based reinforcement learning for generative speech
• Built a KTO pipeline pairing an affect reward model with a duplex audio model's LLM backbone to align generated speech to user affect, using directional (rank-based) rather than pointwise feedback.
• Developed a LoRA fine-tuning method to condition a duplex model's audio output on internal uncertainty signals (semantic entropy, log-probability, hidden-state probes) without added inference latency.
• Ran a between-subjects cross-modality preference study (n=94, 20 prompts each) showing that text and audio judgments of identical semantic content diverge; modeled rater bias with linear mixed-effects models and validated GPT-4o as a modality-dependent human proxy.
• Collected a large-scale LLM-output preference dataset (n≈3,000, 18K+ pairwise judgments via Prolific/Qualtrics) for a study on content-moderation preferences across political affiliation.
• Built a Mixture-of-Experts model with weight averaging, exploring compute/performance trade-offs.

Jan 2022 to Aug 2022

Virufy - *Associate Audio Machine Learning Engineer*
• Fine-tuned audio classifiers via transfer learning (TensorFlow, SK-Learn) for cough-based detection; ran feature/architecture search on SageMaker and peer-reviewed ML research.

July 2021 to Aug 2022

University of California San Diego - *Undergraduate Student Researcher*
• Built logistic-regression and word-embedding models for hate-speech detection on Twitter; scraped data with Selenium; shipped a React/Node/Flask + AWS/SQL tool for non-experts to audit faulty ML systems.

Education

Sep 2022 to June 2027

University of California San Diego
Ph.D. in Computer Science & Engineering

Sep 2019 to June 2021

University of California San Diego
B.S. in Cognitive Science: Machine Learning & Neural Computation; Minor in CSE

Selected Publications (Google Scholar)

Accepted or Published

2. Kaufman et al., WARNING This Contains Misinformation: The Effect of Cognitive Factors, Beliefs, and Personality on Misinformation Warning Tag Attitudes, **CSCW 2025**
1. Kaufman et al., What Did My Car Say? Impact of Autonomous Vehicle Explanation Errors and Driving Context On Comfort, Reliance, Satisfaction, and Driving Confidence, **CHI 2025**

In-prep or Review

6. Broukhim et al., Affect as Implicit Feedback in Speech (in preparation)
5. Broukhim et al., Aligning Perceived Confidence to Generated Speech (in preparation)
4. Broukhim et al., Social Emotion Dysregulation Evaluation Dataset and Models (in preparation)
3. Appel et al., Both Democrats and Republicans Prefer To Receive Multiple Perspectives in LLM Responses to Politically Divisive Topics, (NeurIPS)
2. Broukhim et al., Same Words, Different Judgments: How Preferences Vary Across Modalities, (NeurIPS)
1. Broukhim et al., Preference-Based Learning in Audio Applications: A Systematic Analysis, (CSUR)

Skills

Tech Skills

Languages: Python, SQL, Java, C++, C, JavaScript, HTML/CSS
ML/Frameworks: PyTorch, TensorFlow, Keras, SK-Learn, HuggingFace, NumPy, Pandas, Selenium
Infra/Tools: AWS, SageMaker, Git, Firebase, MySQL, React/Node/Flask